Manufacturing Timeline – Key Points

A comprehensive breakdown of the parametric factors that influence fabrication and assembly time for standard trolley manufacturing, from base assumptions through material and environment considerations.

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Base Time Assumptions & Parametric Scaling Factors

Base Time Assumption

Fabrication time = **5 hours** (baseline)
Assembly time = **2 hours** (baseline)

Represents standard trolley manufacturing effort.

Parametric Time Scaling

Timeline is **not constant** — time increases based on design complexity.

Scaling Factors Overview

Load Factor

Higher load → stronger structure → more work. Increases cutting, welding, and handling time.

Size Factor

Larger platform → more material → more fabrication effort. Affects cutting length and welding length.

Mass Factor

Heavier system → more material processing. Impacts handling and welding effort.

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Configuration, Material & Environment Factors

Wheel Configuration

4 wheels → more assembly effort than 2 wheels.
Additional axle and mounting operations required.

Axle Contribution

Each axle adds extra fabrication and assembly time.
Accounts for machining and fitting effort.

Material Factor

Stainless steel takes more time than mild steel due to: harder welding and additional finishing.

Environment Factor

Outdoor → extra coating/finishing required.
Indoor → simpler processing.

Fabrication Time Includes

- Cutting
- Welding
- Frame preparation
- Surface finishing

Assembly Time Includes

- Wheel mounting
- Axle fitting
- Final inspection

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